

HALo: Hilbert-Space Active Localizability for Scalar Field Navigation

Anonymous Authors

Abstract—Localization in scalar fields is often challenging, especially in a fully sensor-denied environment. Scalar fields produce spatially varying outputs that are gradient-like and often include rich spatial information. However, extracting information from these fields is often realized in rank-deficient formulations due to the single dimensionality. We present Hilbert-Space active localizability for scalar-field navigation, which generates a full-rank localizability or information field for informative navigation. Our method shows a $1.37\times$ improvement over state-of-the-art methods on real scalar fields, with a monotonic decrease in RMSE relative to baseline methods. Code, videos, and datasets are available at <https://halo-nav.github.io>.

I. INTRODUCTION

Global navigation satellite system (GNSS)-denied localization systems remain an open challenge in robotics. Robots frequently operate indoors or in unknown environments that lack accurate localization information, relying on sensing modalities to localize. As the localization through these modalities is often noisy, optimal movement to ensure adequate localizability has become common [1]–[4]. Often, active navigation is computed using information proxies such as filter covariances or entropic values [5], [6]. These proxies are, however, often computationally expensive to evaluate online, as they require propagating full belief distributions over trajectory candidates or relying on assumptions such as Gaussianity or independence of variance, which can limit accuracy in complex environments.

The lack of feature information is particularly pronounced in low-dimensional sensing systems, such as scalar fields. Scalar fields, such as magnetic anomalies, barometric pressure, bathymetry, and terrain elevation, provide persistent but spatially varying information. The low dimensionality of the measurement indicates high spatial variation in localizability [7]. Nonetheless, scalar fields remain heavily leveraged in GNSS-denied navigation due to their passive nature, low power consumption, and reliability in harsh, unstructured environments.

Active navigation in scalar fields is addressed through observability methods [8], greedy-based methods [9], and optimal coverage-based planning [10]. Sampling-based model predictive path integral approaches [11] have been separately extended to incorporate belief or information costs. While these methods provide active formulations for path planning with low-rank information metrics, they tend to focus on pointwise information metrics or on full covariance or belief propagation, which are computationally expensive at the sampling rollout rates required by sampling exploration planners. Furthermore, these planners are generally not equipped

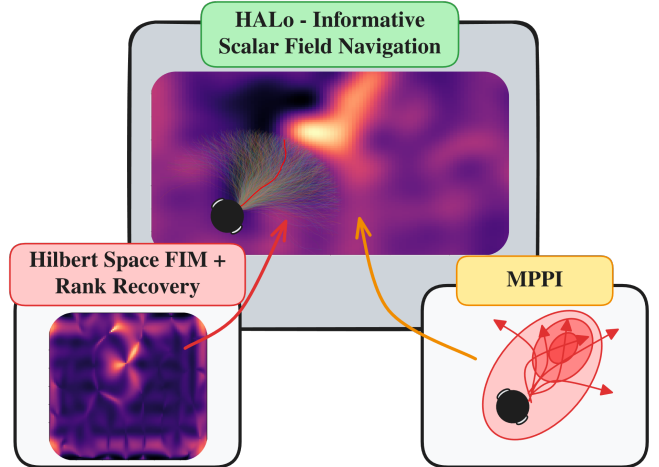


Fig. 1: HALo produces a fast and efficient full-rank Fisher information field for scalar fields that are leveraged through model predictive path integral optimizers for informative path planning. The architecture provides a fully integrated planning method for ensuring scalar fields are utilized.

to handle the rank deficiency of scalar fields, aside from the observability metric.

To address the gap formed by the rank-deficient scalar field localizability, we propose **Hilbert-Space Active Localizability**, a localizability field for scalar field navigation that is leveraged via a sampling planner for rank-deficient exploration. An architectural overview for HALo is given in Fig. 1.

A. Contributions

The following are the main contributions that are presented here:

- 1) **Closed-form Hessian basis for Hilbert-space predicted scalar fields:** We derive the exact Hessian of a Hilbert-space Gaussian process posterior mean in closed form via the eigenbasis’s second-order derivative, extending the closed form gradient computation to full second-order structure.
- 2) **Rank recovery for scalar-field Fisher information:** While the Fisher information indicates localizability, in scalar fields it remains rank deficient due to directional constraints. We recover the rank via a curvature augmentation using eigenbasis Hessian computations.
- 3) **MPPI-based validation of curvature-augmented localizability:** We demonstrate that contributions (1) and (2) make a tractable embedding of the localizability in an MPPI without per-rollout gradient computations, showing improved localization performance.

II. RELATED WORK

Informative Path Planning for Scalar Fields: Information-based path planning is crucial for scalar fields, especially in low-frequency, featureless areas. [10] introduced a sparse Gaussian process (GP) path planning method that exploits covariance-aware regions. [8] shows improvement in leveraging an observability-based system that generates trajectories with weighted gradients. [12] uses entropy approximation methods, though the formulation is highly dependent on the localization filter rather than the environment itself. These methods rely solely on gradient information in scalar fields, which constrains only a single direction in the underlying Fisher information and is therefore generally rank-deficient for $d > 1$. Prior work has explored joint gradient-Hessian estimation through multi-robot formation geometry [13], although not from a rank-recovery perspective.

Gaussian Process Field Representation: GGP field representations for scalar fields are common. [14] and [15] provide a variance-bounded functional representation and yield a calibrated predictor against which information is gathered. [15] presents a Hilbert-space method enabling an $\mathcal{O}(m^3)$ fit with an $\mathcal{O}(m)$ posterior predictor, enabling efficient extraction of information from the field. Sparse GPs provide a similar speed [16], [17] as do Hough-transform Gaussian processes [18] with scaling costs that depend on data size $\mathcal{O}(NM^2)$ in M inducing points. Sparse GPs' adaptive inducing points can allocate resolution toward spatially rich regions; a capability HSGP's fixed domain-spanning basis does not share. We adopt the HSGP specifically because its basis functions are closed-form and data-independent, enabling repeated closed-form evaluations of the gradient and Hessian.

MPPI and Sampling-Based Planning: Model predictive path integral (MPPI) has become a critical addition to sampling-based planning and control, allowing for costs beyond dynamics and terminal objectives. GP-MPPI [19] uses a sparse GP to model navigability uncertainty and bias rollout towards a priori explored regions of free space, which is over the traversability of free space, not the information encoded in the sensing space. PA-MPPI [11] incorporates perception costs directly into the rollout, aligning with the methods presented here. Still, it is posed as a general obstacle and an unknown space-navigation framework for quadrotors, and it does not address issues distinct from the scalar fields presented here, which complicates its applicability. No existing work combines MPPI rollout costs with a physical scalar-field environment for information-aware planning; we have introduced a framework to address the scalar-field-specific issues this setting raises.

III. PROBLEM FORMULATION

A. Workspace and Robot Frames

The workspace is defined by a 2-dimensional space $\mathcal{W} \subset \mathbb{R}^2$. The inertial frame $\mathcal{F}_W$ with x_I, y_I, z_I -axes is defined in $\mathcal{W}$, and the body-fixed frame $\mathcal{F}_B$ with x_B, y_B, z_B -axes is defined and fixed on the mobile robot. The robot position

and orientation are defined by the position and orientation of $\mathcal{F}_B$ concerning $\mathcal{F}_W$.

B. Motion Model and Localization

Localization is performed solely on the onboard measurements and a given scalar field map, without other sensors such as an internal measurement unit. In this paper, we assume that the mobile robot is a differential-drive ground robot, although this assumption can be easily generalized to account for other dynamic constraints. Then, the position of the robot is defined by x and y position of $\mathcal{F}_B$ in $x_I y_I$ -plane, and the orientation of the robot is defined by θ , which is the angle of x_B -axis concerning x_I -axis. By denoting a certain time step by $k = 0, 1, 2, \dots$, a state vector that represents the robot state at time k is defined by $\mathbf{x}_k = [x_k \ y_k \ \theta_k \ V_k]^\top$.

The control state vector at time step k is specified by $\mathbf{u}_k$, the robot's motion model is:

$$\mathbf{x}_{k+1} = F(\mathbf{x}_k, \mathbf{u}_k) \quad (1)$$

The motion model of a differential-drive robot can be derived from a unicycle model. By letting a and ω be the linear acceleration and angular velocity, the kinematics of a unicycle model gives

$$\begin{aligned} x_{k+1} &= x_k + V_k \cos \theta \Delta t + \epsilon_x \\ y_{k+1} &= y_k + V_k \sin \theta \Delta t + \epsilon_y \\ \theta_{k+1} &= \theta_k + \omega \Delta t + \epsilon_\theta \\ V_{k+1} &= V_k + a \Delta t + \epsilon_v \end{aligned} \quad (2)$$

where Δt is the size of each time step and $\epsilon_x, \epsilon_y, \epsilon_\theta, \epsilon_v$ are random Gaussian noises with zero mean and standard deviations $\sigma_x, \sigma_y, \sigma_\theta$, and σ_v , respectively. The robot's control input is defined by $\mathbf{u}_k = [a \ \omega]^\top$.

The sensor measurement model describes how scalar-field measurements are injected into the state estimation filter. The scalar-field reading obtained from the state $\mathbf{x}_k$ is denoted by $z_k \in \mathbb{R}^+$.

$$z_k = f(\mathbf{x}_k) + \epsilon_z, \quad \epsilon_z \sim \mathcal{N}(0, \sigma_R) \quad (3)$$

where f is a functional representation of the scalar field and σ_R is the standard deviation of the measured scalar field. The robot's pose was determined at each time step using a particle filter following [20].

Problem statement: Given a scalar field model $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ learned in closed form using a sparse set of scalar field measurements $\mathbf{Y} \in \{y_1, y_2, \dots, y_n\}$ collected at input positions $\mathbf{X} \in \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ over a workspace $\mathcal{W} \subset \mathbb{R}^2$. The goal is to plan a control sequence $u_{0:T-1}$ over a receding horizon that jointly (i) drives the robot to a goal state and (ii) maximizes the localizability of the robot's pose estimate, quantified by the smallest eigenvalue of the Fisher information matrix $I(\mathbf{x})$ accumulated along a candidate trajectory. The pointwise Fisher information matrix in scalar fields, however, is rank-deficient, leading to uninformative directions in the gradients. The overall goal is to minimize

a trajectory optimization objective:

$$\mathbf{u}_{0:T-1}^* = \arg \min_{\mathbf{u}_{0:T-1}} \mathbb{E} \left[\phi(\mathbf{x}_T) + \sum_{t=0}^{T-1} c(\mathbf{x}_t, \mathbf{u}_t) \right] \quad (4)$$

subject to (1), where $c(\cdot)$ is the cost function defined in Sec. V-C. $\mathbf{u}_0^*$ is applied before re-planning.

IV. GAUSSIAN PROCESS PRELIMINARIES

The canonical form of Gaussian process regression, model functions $f(\mathbf{x})$ are assumed to be realizations from a Gaussian random process prior with a given covariance function $\kappa(\mathbf{x}, \mathbf{x}')$. Learning is finding the posterior process at a desired input $\mathbf{x}_*$ given a set of noisy measurements $\mathbf{Y} \in \{y_1, y_2, \dots, y_n\}$ observed at inputs $\mathbf{X} \in \{\mathbf{x}_1, \mathbf{x}_2, \dots\}$. The prior model is written in the form:

$$\begin{aligned} f(\mathbf{x}) &\sim \mathcal{GP}(0, \kappa(\mathbf{x}, \mathbf{x}')) \\ y_i &= f(\mathbf{x}_i) + \epsilon_i \end{aligned} \quad (5)$$

where y_i is corrupted by Gaussian noise $\epsilon_i \sim \mathcal{N}(0, \sigma_{\text{noise}}^2)$, for $i = 1, 2, \dots, N$. Prediction for unseen inputs is based on the conditional posterior probability. The conditional mean and variance are:

$$\begin{aligned} \mathbb{E}[f(\mathbf{x}_*)] &= \mathbf{k}_*^\top (\mathbf{K} + \sigma_{\text{noise}}^2 \mathbf{I}_n)^{-1} \mathbf{Y} \\ \mathbb{V}[f(\mathbf{x}_*)] &= \kappa(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{k}_*^\top (\mathbf{K} + \sigma_{\text{noise}}^2 \mathbf{I}_n)^{-1} \mathbf{k}_* \end{aligned} \quad (6)$$

where $\mathbf{K}_{i,j} = \kappa(\mathbf{x}_i, \mathbf{x}_j)$, $\mathbf{k}_*$ is an n dimensional vector with the i th entry being $\kappa(\mathbf{x}_*, \mathbf{x}_i)$.

A. Reduced Rank Hilbert Space GP

The primary issue with the closed-form conditional mean and variance is the inversion of the Gram matrix, an $N \times N$ matrix $\mathbf{K} + \sigma_{\text{noise}}^2 \mathbf{I}_n$, which would be an $\mathcal{O}(N^3)$ operation. [21] shows that the inversion is accelerated via the matrix inversion lemma [22], thereby speeding up the computation.

The approximate Gram matrix $\tilde{\mathbf{K}}$ optimal reduced rank formula is $\tilde{\mathbf{K}} = \Phi \Lambda \Phi^\top$, where Φ is the matrix of orthonormal eigen vectors and Λ is the corresponding matrix of diagonal eigenvalues. However, computing the eigendecomposition is expensive ($\mathcal{O}(m^3)$). [15] showed that this eigendecomposition can be approximated through the expansion of the Laplace operator (∇^2) in a compact domain Ω . They interpret the covariance function $\kappa(\mathbf{x}, \mathbf{x}')$ as a pseudo-differential operator $\mathcal{K}$ to facilitate a Hilbert space approximation.

Let $\Omega \subset \mathbb{R}^d$ be the compact set and review the eigenvalue problem for Laplace operators with boundary conditions (Dirichlet boundary conditions).

$$\begin{cases} -\nabla^2 \phi_j(\mathbf{x}) = \lambda_j \phi_j(\mathbf{x}), & \mathbf{x} \in \Omega \\ \phi_j(\mathbf{x}) = 0, & \mathbf{x} \in \partial\Omega \end{cases} \quad (7)$$

where the boundary $\partial\Omega$ is sufficiently smooth, to ensure the eigenvalues exist. [15] shows that under the Dirichlet boundary conditions on the compact domain defined, an isotropic covariance function admits the spectral approximation:

$$\kappa(\mathbf{x}, \mathbf{x}') \approx \sum_j S(\sqrt{\lambda_j}) \phi_j(\mathbf{x}) \phi_j(\mathbf{x}') \quad (8)$$

where ϕ_j, λ_j solve (7), and $S(\cdot)$ is the spectral density of the covariance function. This solution is generally valid away from the boundary, tending to deteriorate around $\partial\Omega$ as defined by (7).

Now that the kernel approximation has been applied, the GP regression can be formulated as in (5). The approximation simplifies the $n \times n$ matrix inversion by projecting the process onto a truncated set of m Laplacian basis functions. $f(\mathbf{x}) \approx \sum_{j=1}^m f_j \phi_j(\mathbf{x})$ where $f_j \sim \mathcal{N}(0, S(\sqrt{\lambda_j}))$

Then the eigen approximation of the Gram matrix becomes $\mathbf{K} \approx \Phi \Lambda \Phi^\top$, where Λ is a diagonal matrix of the leading m approximate eigen values $\Lambda_{jj} = S(\lambda_j)$, $j = 1, 2, \dots, m$. The corresponding eigenvectors are given by $\Phi_{ij} = \phi_j(\mathbf{x}_i)$.

Through the matrix inversion lemma [22], the conditional mean and variance are rewritten in terms of the spectral eigen decomposition.

$$\begin{aligned} \mathbb{E}[f(\mathbf{x}_*)] &= \phi_*^\top (\Phi^\top \Phi + \sigma_{\text{noise}}^2 \Lambda^{-1})^{-1} \Phi^\top \mathbf{Y} \\ \mathbb{V}[f(\mathbf{x}_*)] &= \sigma_N^2 \phi_*^\top (\Phi^\top \Phi + \sigma_{\text{noise}}^2 \Lambda^{-1})^{-1} \phi_* \end{aligned} \quad (9)$$

where ϕ_* is an m -dimensional vector with the j^{th} entry being $\phi_j(\mathbf{x}_*)$. This spectral approximation with a reduced basis set does scale in accuracy as the basis set size increases ($m \rightarrow n$), but the computational complexity scales linearly with m .

V. PROPOSED METHODOLOGY

The proposed method, which replaces the original entropy map, is a two-field solution. First, a field representing the Fisher information (how useful the sensor reading is) and an ambiguity field (how globally unique the area is) are extracted. Both of these fields complement each other and are necessary due to the scalar field localization being performed, with the promise of scaling up to multimodal perceptual sensing as well.

A. Fisher Information Field

The Fisher information matrix (FIM) quantifies how sensitive the likelihood is to the unknown parameters. The FIM is defined as the expectation of the outer product of the score function:

$$I_{\mathbf{x}}(z) = \mathbb{E}_{\mathbf{z}|\mathbf{x}} \left[\frac{\partial^2 (-\ln p(\mathbf{z} | \mathbf{x}))}{\partial \mathbf{x} \partial \mathbf{x}^\top} \right] \quad (10)$$

Alternatively, if the GP defines the measurement model in (5), where z is corrupted by zero-mean Gaussian noise, the FIM can be written as:

$$I_{\mathbf{x}}(z) = \frac{1}{\sigma^2} J_{\mathbf{x}} J_{\mathbf{x}}^\top, \quad J_{\mathbf{x}} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}} \quad (11)$$

The inverse Fisher information bounds the covariance of any unbiased estimate $\hat{\mathbf{x}}$, based on the measurement $\mathbf{z}$.

$$\text{Cov}(\hat{\mathbf{x}}) = \mathbb{E}[(\hat{\mathbf{x}} - \mathbf{x})^\top (\hat{\mathbf{x}} - \mathbf{x})] \geq I_{\mathbf{x}}^{-1} \quad (12)$$

This bounding is commonly known as the Cramér-Rao lower bound (CRLB). Substituting in equations from the GP with

the mean and variance on $f(\mathbf{x})$ denoted as $\mu_f(\mathbf{x})$ and $\sigma_f^2(\mathbf{x})$, we get:

$$I_{\mathbf{x}} = \frac{1}{\sigma_{\text{noise}}^2 + \sigma_f^2} \nabla \mu(\mathbf{x}) \nabla \mu(\mathbf{x})^\top \quad (13)$$

which is a single-point estimate of the FIM. If a field is to be defined, the FIM can be aggregated. [5] aggregated these values over visible landmarks for vision-based Fisher information fields (FIF) as:

$$I_{T_i} = \sum_k^{k \in V_i} I_{T_i}(p_k^w) \quad (14)$$

where V_i is the index set of landmarks, p_k^w is the k th landmark in the world frame and T_i is the i th sample pose. Similarly, the Fisher information for scalar fields can be aggregated over a sampled trajectory $\mathcal{Q}$ by leveraging the previously defined HSGP machinery.

$$I_i \equiv I_{\mathbf{x}}(\mathbf{x}_{*,i}) = \frac{1}{\sigma_n^2 + \sigma_f^2(\mathbf{x}_{*,i})} \nabla \mu(\mathbf{x}_{*,i}) \nabla \mu(\mathbf{x}_{*,i})^\top \quad (15)$$

where $P \in \mathbb{N}$ is the number of query locations within the neighborhood and $\mathbf{x}_{*,i} \in \mathcal{Q}$, $i = 1, \dots, P$.

The quantity I_i is rank-deficient, as it is formed from the outer product of the gradient of a scalar-valued field with respect to a d -dimensional position. Consequently, $\text{rank}(I_i) \leq 1$. A scalar measurement provides local information only along the field gradient direction, while motion tangent to an equal-valued contour is locally unobservable. The FIM only characterizes local sensitivity, ignoring global ambiguities that arise when spatially separated locations yield similar observations. We therefore introduce a curvature-based rank recovery to complement the FIM for scalar fields.

B. Rank Recovery via Local Curvature

Expanding the gradient about a query point $\mathbf{x}_0$ to first order through random perturbations,

$$\nabla f(\mathbf{x}_0 + \delta) \approx \nabla f(\mathbf{x}_0) + H(\mathbf{x}_0)\delta \quad (16)$$

where $H(\mathbf{x}_0) \in \mathbb{R}^{d \times d}$ is the Hessian of f at $\mathbf{x}_0$. Summing outer products of $\nabla f(\mathbf{x}_0 + \delta_i)$ over P offsets δ_i drawn symmetrically about $\mathbf{x}_0$ (zero first moment), the cross terms vanish and the gradient outer product simplifies:

$$\begin{aligned} & \sum_{i=1}^P \nabla f(\mathbf{x}_0 + \delta_i) \nabla f(\mathbf{x}_0 + \delta_i)^\top \\ & \approx P \nabla f(\mathbf{x}_0) \nabla f(\mathbf{x}_0)^\top + H(\mathbf{x}_0) M H(\mathbf{x}_0)^\top \end{aligned} \quad (17)$$

where $M = \sum_i \delta_i \delta_i^\top$ is the second moment of the offset pattern (scatter matrix). The discrete trajectory aggregation $I_{\mathcal{Q}}$ of Sec. V-A is thus a sampled realization of this identity; in the limit of a continuum set, it converges to a closed-form correction term $H(\mathbf{x}) M H(\mathbf{x})^\top$, which we adopt directly rather than sampling.

Proposition 1 (Rank recovery): Let the direction constrained by the pointwise FIM be denoted as $\mathbf{v} = \nabla f(\mathbf{x}_0) / \|\nabla f(\mathbf{x}_0)\|$. The aggregated FIM $I_{\mathcal{Q}}(\mathbf{x}_0) \approx P \nabla f \nabla f^\top + H M H^\top$ is full rank if and only if (i) M is full

rank, and (ii) $H(\mathbf{x}_0)$ is non-degenerate on the orthogonal complement of $\mathbf{v}$. Condition (i) is a design choice: an isotropic offset pattern (or, in the closed-form limit, $M \propto I_d$) guarantees full rank in M . Condition (ii) is a property of the field itself. If the local curvature is deficient in a direction orthogonal to $\mathbf{v}$ across the domain, no patch design recovers observability in that direction.

For the separable box-domain eigenbasis $\phi_j(\mathbf{x}) = \prod_k \phi_{j,k}(x_k)$, the Hessian basis follows from differentiating the per-dimension sine factors:

$$\begin{aligned} \frac{\partial^2 \phi_j}{\partial x_k^2} &= -\lambda_{j,k} \phi_j(\mathbf{x}), \\ \frac{\partial^2 \phi_j}{\partial x_k \partial x_l} \Big|_{k \neq l} &= \sqrt{\lambda_{j,k} \lambda_{j,l}} \psi_{j,k}(x_k) \psi_{j,l}(x_l) \prod_{d \neq k,l} \phi_{j,d}(x_d) \end{aligned} \quad (18)$$

where $\lambda_{j,k}$ is the per-dimension eigenvalue contribution ($\lambda_j = \sum_k \lambda_{j,k}$) and $\psi_{j,k}$ is the cosine partner of $\phi_{j,k}$. The diagonal follows from the definition of the per-dimension Laplacian eigenvalue problem. At the same time, the off-diagonals arise from the product rule applied to the mixed partial derivative of the separable product, since each factor depends on a single coordinate; for example, differentiating once on x_k and once in x_l replaces the factors with their first derivatives, respectively. Tightening against the posterior coefficients,

$$H(\mathbf{x}) = \sum_j w_j H_j(\mathbf{x}), \quad [H_j(\mathbf{x})]_{kl} = \frac{\partial^2 \phi_j}{\partial x_k \partial x_l}(\mathbf{x}) \quad (19)$$

gives the exact Hessian of the posterior mean at any query point, in closed form, consistent with the gradient and value computations above using the eigenbasis, providing efficient $\mathcal{O}(Pmd^2)$ computations per rollout after an initial $\mathcal{O}(m^3)$ computation. Combining Prop. 1 with the point-estimate FIM, the localizability field is defined as

$$I(\mathbf{x}) = \mathbb{E}[I_{\mathbf{x}}] + \gamma H(\mathbf{x}) H(\mathbf{x})^\top \quad (20)$$

where γ absorbs the patch second-moment scale M . The scalar localizability score used by the planner is the smallest eigenvalue of $I(\mathbf{x})$, corresponding to the worst-constrained direction under the Cramér-Rao bound:

$$\ell(\mathbf{x}) = \lambda_{\min}(I(\mathbf{x})) \quad (21)$$

C. Model Predictive Path Integral

The model predictive path integral setup follows from relevant literature [11], [23], [24]. The control update at each iteration is given by an importance-sampling-weighted average of K sampled control perturbations. The updated law is given:

$$w_k = \frac{\exp(-\frac{1}{\lambda}(C(\tau_k) - \min_j C(\tau_j)))}{\sum_{k=1}^K \exp(-\frac{1}{\lambda}(C(\tau_k) - \min_j C(\tau_k)))} \quad (22)$$

where each candidate control is given by $\mathbf{u}_t^* = \bar{\mathbf{u}}_t + \sum_{k=1}^K w_k \epsilon_{t,k}$ with τ_k is the k -th sampled trajectory rollout, $C(\cdot)$ is the total cost function and λ is the temperature normalization for the softmax weighting defined by w_k

1) *Cost Function Design*: The cost function for the MPPI trajectory rollout τ_k is a common additive function that includes goal, velocity, control, localizability, and terminal costs. The running goal cost c_{goal} is defined as the quadratic tracking error to the goal position accumulated over the horizon:

$$c_{\text{goal}}(\tau_k) = \sum_{t=0}^{T-1} \|\mathbf{p}_t - \mathbf{p}_{\text{goal}}\|^2 \quad (23)$$

where $\mathbf{p}_t$ is the coordinates of the rollout at t and $\mathbf{p}_{\text{goal}}$ is the coordinates of the goal. While c_{goal} pulls the trajectories towards the goal, the MPPI can overshoot due to high velocity states near the goal. To counterbalance this, a running velocity cost that penalizes speed near the goal is defined based on a distance $d_t = \|\mathbf{p}_t - \mathbf{p}_{\text{goal}}\|$:

$$c_{\text{vel}}(\tau_k) = \sum_{t=0}^{T-1} \exp(-d_t) V_t^2 \quad (24)$$

The control penalty to ensure the controls are not infeasible are:

$$c_{\mathbf{u}}(\tau_k) = \sum_{t=0}^{T-1} \|\mathbf{u}_t\|^2 \quad (25)$$

The localizability costs, which are derived from the rank-recovered Fisher information matrix described in Sec. V-A and V-B, are described as:

$$c_{\text{loc}}(\tau_k) = - \sum_{t=0}^{T-1} \ell(\mathbf{p}_t) \tanh\left(\frac{d_t}{0.5}\right) \quad (26)$$

where $\ell(\mathbf{x})$ is given in (21). Finally, a terminal cost is designed to penalize the trajectories that end farther from the goal as:

$$c_{\text{terminal}}(\tau_k) = \|\mathbf{p}_T - \mathbf{p}_{\text{goal}}\|^2 + w_{\text{vel}} V_T^2 \quad (27)$$

Each of these final costs is traditionally combined without normalization. However, due to the localizability costs being dependent on the scalar field, the values output by ℓ are vastly different. Therefore, a z -score normalization is applied to ensure all costs fall within the same range.

$$\hat{c}(\tau_k) = \frac{c(\tau_k) - \mu_K(c)}{\sigma_K(c) + \epsilon} \quad (28)$$

The overall cost function is defined as:

$$C(\tau_k) = w_{\text{loc}} \hat{c}_{\text{loc}} + \hat{c}_{\text{goal}} + w_{\text{vel}} \hat{c}_{\text{vel}} + w_{\mathbf{u}} \hat{c}_{\mathbf{u}} + w_{\text{terminal}} \hat{c}_{\text{terminal}} \quad (29)$$

which is used to select the best control for each step k .

VI. EXPERIMENTAL SETUP

Two sets of experiments were conducted: a simulated map experiment used to validate the localizability term, and a real scalar field experiment using a magnetic anomaly field to compare against state-of-the-art scalar-field and information-driven path-planning approaches.

A. Simulation Setup

The simulation experiments used an artificial scalar field with a single high-amplitude peak. The peak was placed in the middle of the map, and the robot was tasked with moving nearby in low spatially varying areas. The MPPI has a horizon of 30, 2000 samples, and $\Delta t = 0.1$. The particle filter was initialized with a Gaussian centered at the robot's start position with a high variance and a random shift to generate Monte Carlo-style statistical validation. The particle filter performance was evaluated on the root mean square error (RMSE) as $\text{RMSE} = \frac{\|\mathbf{x}_{\text{true}} - \mathbf{x}_{\text{est}}\|}{\sqrt{d}}$.

Simulations were conducted on a desktop with a 12th-generation i7 processor and an NVIDIA 3070 Ti GPU. The GPU-accelerated GP training, though the reduced-rank approximation allows for real-time CPU-only operation. The GPU was solely for efficient gradient computations and vectorization of the MPPI via CUDA-based JAX JIT compilations.

B. Real Scalar Field Experiments

The experiments were run on a physical Clearpath Robotics TurtleBot4 and in simulation. The room is set up with an Optitrack motion capture system for ground-truth positioning. The robot has an onboard Raspberry Pi 4 compute module and an NVIDIA Jetson Orin Nano. The simulation was primarily run for the MPPI-based systems, as the system could not run on the hardware due to an incompatible optimization issue and inefficient rollouts.

The robot is equipped with a quantum hybrid scalar/vector magnetometer from QUSpin. The vector input is disregarded to ensure consistency with the scalar-field formulation. Magnetic anomalies tend to be fairly stable and allow for statistical validation on a temporally stable map that is minimally affected by outside sources.

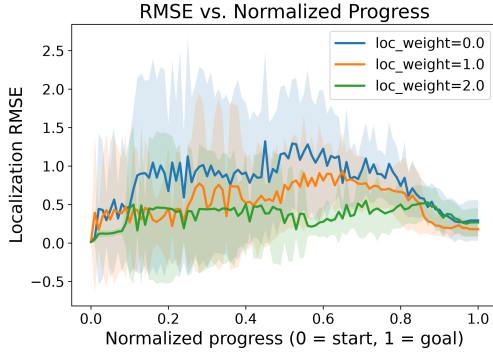
HALo was compared against: (i) *InfoMagNav* [12] an entropy guided information-driven guidance framework using particle filters, (ii) *E-RRT* and exploration guided sampling planner [25], (iii) *InfoGainGreedy*, a greedy formulation for information gain exploration in the map using entropy, (iv) *MPPI*, which has no information cost associated with it, and (v) *FIM-MPPI*, an MPPI formulation that uses Fisher information matrix with its rank deficiency.

VII. RESULTS AND DISCUSSION

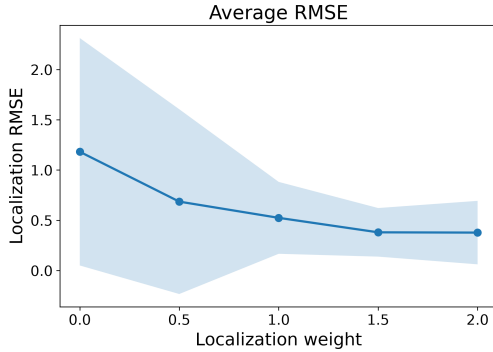
A. Simulation

The simulation results, conducted on a scalar field with a single high-amplitude peak, show that increasing the Fisher-information weight biases the trajectory toward the spatially varying region surrounding the peak, improving localization performance relative to the unweighted baseline. The weight was swept from 0.0 to 2.0 across $n = 10$ matched-seed trials per condition; results are shown in Fig. 2 and 3 and Table I.

Fig. 2(a) shows localization RMSE over normalized trajectory progress. The baseline shows the highest mean RMSE and variance, with a pronounced error spike mid-trajectory that is not observed at any nonzero weight. Goal-reachability improves with information weighting, from 70% to 90%



(a)



(b)

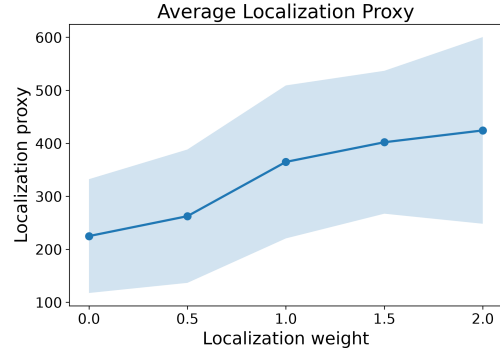
Fig. 2: Simulation results of HALo. (a) shows the RMSE versus normalized progress along a trajectory averaged over $n = 10$ trials. (b) Shows the average RMSE over a full trajectory versus the localizability weighting in the MPPI cost function.

TABLE I: Weight-sweep results, $n = 10$ matched seeds per condition. p -values are paired Wilcoxon signed-rank vs. $\text{loc_weight} = 0.0$ on rmse_average .

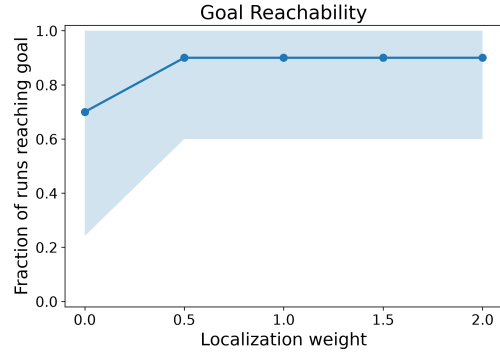
Localizability Weight	Avg. RMSE	Goal Reached	p vs. baseline
0.0	1.18	70%	—
0.5	0.69	90%	0.014
1.0	0.52	90%	0.131
1.5	0.38	90%	0.049
2.0	0.38	90%	0.106

across all nonzero weights. Fig. 2(b) shows the average RMSE over the entire snapshot compared against localizability weights. The localization RMSE monotonically decreases with increasing weight, including a decrease in the variance of the error, displaying consistent results.

Fig. 3(a) shows the weight-sweep average of trajectory RMSE alongside the localizability proxy, which increases monotonically with weight, confirming the planner biases trajectories toward higher information as intended. Average trajectory RMSE decreases monotonically with weight increase, from 1.18 at $\text{loc_weight} = 0.0$ to 0.38 at $\text{loc_weight} = 1.5\text{--}2.0$, with the reduction reaching statistical significance at two out of four weights (paired Wilcoxon signed-rank; see Table I).



(a)



(b)

Fig. 3: Simulation results of HALo. The localizability weight was increased to more strongly influence the navigation algorithm and evaluate field performance using (a) a proxy for localization and (b) goal reachability.

B. Real Scalar Field

HALo was evaluated on a real scalar field collected in an indoor setting with a quantum magnetometer. HALo and the MPPI baselines conclusively outperform the comparative algorithms in both the average and final RMSE. The algorithms are also fairly efficient in terms of path length, with standard deviations from the diagonal test path.

HALo performs better than the baselines, with an average RMSE of 0.85 ± 0.017 , yielding more consistent results than the other formulations. Similarly, its path efficiency exceeds that of MPPI+FIM and MPPI, with MPPI exhibiting significant deviations to achieve results comparable to HALo, highlighting the advantage of the rank-recovery mechanism proposed in this paper.

VIII. CONCLUSIONS

We presented HALo, a Hilbert-space-based active localizability framework for scalar-field navigation. HALo derives the full rank closed-form Hessian of the HSGP posterior mean via the eigenbasis’s second-order derivatives, enabling efficient curvature-augmented rank recovery of the Fisher information matrix. The resulting full-rank localizability field is embedded as a rollout cost in an MPPI planner, providing information-aware trajectory optimization without per-rollout gradient evaluations. Simulation experiments confirm that increasing the localizability weight monotonically improves

TABLE II: Navigation results. All methods navigate from $(-2.5, -1.0)$ to $(2.5, 1.5)$ m across a real magnetic scalar field. RMSE is localization RMSE (particle-filter pose estimate vs. Mocap ground truth). Path efficiency is path length divided by straight-line distance. Goal reached uses a 0.2 m radius criterion. The MPPI, MPPI+FIM, and HALo were only evaluated on the real scalar field map in simulation, not on hardware.

Method	N	RMSE _{avg} (m)	RMSE _{final} (m)	Path Eff.
E-RRT	5	0.623 ± 0.305	1.078 ± 0.648	0.741 ± 0.084
InfoGainGreedy	4	0.681 ± 0.248	1.605 ± 0.639	0.988 ± 0.006
InfoMagNav	5	0.350 ± 0.110	0.534 ± 0.320	0.991 ± 0.005
MPPI	5	0.108 ± 0.021	0.078 ± 0.048	0.918 ± 0.079
MPPI+FIM	5	0.117 ± 0.017	0.145 ± 0.085	0.911 ± 0.054
HALo	5	0.085 ± 0.017	0.059 ± 0.030	0.928 ± 0.71

particle-filter RMSE and goal-reaching success from 70% to 90%. Experiments on a real magnetic anomaly field demonstrate that HALo achieves more efficient average and final localization RMSE than gradient-only, entropy-based, and rank-deficient FIM baselines, validating the practical benefit of curvature-based rank recovery for scalar field navigation.

A. Limitations

HALo inherits many limitations of its underlying components. First, the HSGP approximation uses a fixed-domain basis that does not adapt to the spatial resolution of locally rich field structures. Second, the Fisher information derived here is a globally informed local metric, masking the prior eigenbasis information that the HSGP exposes. This limitation is marked as future work, as the rank-deficient formulation was in part derived as a larger ambiguity metric. Third, the MPPI-based trajectory planner could not be fully evaluated on hardware due to real-time feasibility constraints, limiting direct physical comparison against baseline methods. Future work will address these limitations to improve the effectiveness of the proposed field and expand it to other modalities.

REFERENCES

- [1] C. Stachniss, G. Grisetti, and W. Burgard, "Information Gain-based Exploration Using Rao-Blackwellized Particle Filters," in *Robotics: Science and Systems I*. Robotics: Science and Systems Foundation, Jun. 2005.
- [2] B. Mu *et al.*, "Information-based Active SLAM via topological feature graphs," in *2016 IEEE 55th Conference on Decision and Control (CDC)*, Dec. 2016, pp. 5583–5590.
- [3] W. Burgard, D. Fox, and S. Thrun, "Active Mobile Robot Localization," Aug. 1997.
- [4] M. F. Ahmed *et al.*, "Active SLAM: A Review on Last Decade," *Sensors*, vol. 23, no. 19, p. 8097, Jan. 2023.
- [5] Z. Zhang and D. Scaramuzza, "Beyond Point Clouds: Fisher Information Field for Active Visual Localization," in *2019 International Conference on Robotics and Automation (ICRA)*, May 2019, pp. 5986–5992, iSSN: 2577-087X.
- [6] T. Khatib *et al.*, "Robotic Information Gathering via Deep Generative Inpainting," in *Proc. IEEE Int. Conf. Syst. Man Cybern.*, Oct. 2023, pp. 3130–3137.
- [7] R. McDonald, "Mission-Oriented Multirobot Adaptive Navigation of Scalar Fields," *Engineering Ph.D. Theses*, Feb. 2020.
- [8] J. H. Ramos *et al.*, "Information-Aware Guidance for Magnetic Anomaly based Navigation," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.* Kyoto, Japan: IEEE, 2022, pp. 6347–6354.
- [9] M. Popović *et al.*, "Informative Path Planning for Active Field Mapping under Localization Uncertainty," in *2020 IEEE International Conference on Robotics and Automation (ICRA)*, May 2020, pp. 10 751–10 757, iSSN: 2577-087X.
- [10] K. Jakkala *et al.*, "Informative Path Planning with Guaranteed Estimation Uncertainty," vol. 22, Jul. 2026.
- [11] Y. Zhai, R. Reiter, and D. Scaramuzza, "PA-MPPI: Perception-Aware Model Predictive Path Integral Control for Quadrotor Navigation in Unknown Environments," *IEEE Robotics and Automation Letters*, vol. 11, no. 3, pp. 3804–3811, Mar. 2026.
- [12] A. Penumarti *et al.*, "Real-Time Uncertainty-Aware Motion Planning for Magnetic-Based Navigation," *IEEE Access*, vol. 14, pp. 23 824–23 834, 2026.
- [13] L. Briñón-Arranz, A. Renzaglia, and L. Schenato, "Multirobot Symmetric Formations for Gradient and Hessian Estimation With Application to Source Seeking," *IEEE Transactions on Robotics*, vol. 35, no. 3, pp. 782–789, Jun. 2019.
- [14] M. Kok and A. Solin, "Scalable Magnetic Field SLAM in 3D Using Gaussian Process Maps," in *Proc. Int. Conf. Inf. Fusion*. Cambridge, United Kingdom: IEEE, Jul. 2018, pp. 1353–1360.
- [15] A. Solin and S. Särkkä, "Hilbert space methods for reduced-rank Gaussian process regression," *Statistics and Computing*, vol. 30, no. 2, pp. 419–446, Mar. 2020.
- [16] K. Jakkala and S. Akella, "Multi-Robot Informative Path Planning from Regression with Sparse Gaussian Processes," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, May 2024, pp. 12 382–12 388.
- [17] K. Jakkala and S. Akella, "Fully Differentiable Adaptive Informative Path Planning," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, May 2025, pp. 5431–5437.
- [18] M. Qureshi *et al.*, "Scalar Field Mapping with Adaptive High-Intensity Region Avoidance," in *2024 IEEE Conference on Control Technology and Applications (CCTA)*, Aug. 2024, pp. 388–393, iSSN: 2768-0770.
- [19] I. S. Mohamed, M. Ali, and L. Liu, "GP-Guided MPPI for Efficient Navigation in Complex Unknown Cluttered Environments," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Oct. 2023, pp. 7463–7470, iSSN: 2153-0866.
- [20] S. Thrun, W. Burgard, and D. Fox, *Probabilistic Robotics*, ser. Intelligent Robotics and Autonomous Agents series. The MIT Press, Aug. 2005.
- [21] C. E. Rasmussen and C. K. I. Williams, *Gaussian processes for machine learning*, 3rd ed., ser. Adaptive computation and machine learning. Cambridge, Mass.: MIT Press, 2008.
- [22] M. A. Woodbury, *Inverting Modified Matrices*. Department of Statistics, Princeton University, 1950.
- [23] G. Williams *et al.*, "Aggressive driving with model predictive path integral control," in *2016 IEEE International Conference on Robotics and Automation (ICRA)*, May 2016, pp. 1433–1440.
- [24] G. Williams *et al.*, "Information-Theoretic Model Predictive Control: Theory and Applications to Autonomous Driving," *IEEE Transactions on Robotics*, vol. 34, no. 6, pp. 1603–1622, Dec. 2018.
- [25] B. Lindqvist, A.-A. Agha-Mohammadi, and G. Nikolakopoulos, "Exploration-RRT: A multi-objective Path Planning and Exploration Framework for Unknown and Unstructured Environments," in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Sep. 2021, pp. 3429–3435.